ICAO Doc 9303 Appendix: Check Digit Calculation Rules for MRZ Zones

Type: technical_guide | **Date:** 2020-03-01

1. Weighting Factors and Modulo 10 Algorithm

The check digit calculation uses a repeating sequence of weighting factors: 7, 3, 1, 7, 3, 1... applied sequentially from left to right across all characters in the data element. The numerical value of each character is multiplied by its corresponding weight. The sum of all products is divided by 10; the remainder is the check digit.

2. Character Value Mapping Table

The character set consists of digits 0-9, capital letters A-Z, and the filler character '<'. Digits 0 through 9 have values 0 through 9. Letters A through Z have values 10 through 35 respectively (A=10, B=11 ... Z=35). The filler character '<' is assigned a numeric value of 0.

3. Composite Check Digit Calculation

The composite check digit in TD3 line 2 (position 44) verifies the combined integrity of the passport number + its check digit, the date of birth + its check digit, and the date of expiry + its check digit, plus any optional data and its check digit. A failure in the composite check indicates OCR misread or physical alteration.